

Lecture Notes: Logic

1. Introduction to Logic

Logic is the study of principles of reasoning and argumentation. It provides formal methods for distinguishing valid reasoning from fallacious arguments and is fundamental in mathematics, computer science, and philosophy.

Types of Logic

1. **Propositional Logic:** Deals with propositions (statements that are either true or false) and logical connectives such as **AND**, **OR**, and **NOT**.
 2. **First-Order Logic (Predicate Logic):** Extends propositional logic by including quantifiers ($\forall, \exists$) and predicates that express relationships between objects.
-

2. Propositional and First-Order Logic

Propositional Logic

A proposition is a declarative statement that can be assigned a truth value (**true** or **false**). Logical connectives allow us to construct compound statements:

- **Negation** ($\neg P$): The opposite of P .
- **Conjunction** ($P \wedge Q$): True if both P and Q are true.
- **Disjunction** ($P \vee Q$): True if at least one of P or Q is true.
- **Implication** ($P \rightarrow Q$): True unless P is true and Q is false.
- **Biconditional** ($P \leftrightarrow Q$): True if P and Q have the same truth value.

First-Order Logic (FOL)

While propositional logic deals with whole statements, **first-order logic** allows us to express statements about objects and their properties using:

- **Quantifiers:**
- **Universal quantifier** ($\forall x$): “For all x , some property holds.”
- **Existential quantifier** ($\exists x$): “There exists an x for which a property holds.”
- **Predicates:** Functions that return true or false for given objects (e.g., $P(x)$ could mean “ x is a prime number”).

Example of a formalized statement in FOL:

$$\forall x (Human(x) \rightarrow Mortal(x))$$

(“All humans are mortal.”)

3. Formalization of Arguments

Logical Validity

An argument consists of premises and a conclusion. It is **valid** if, whenever the premises are true, the conclusion must also be true.

Common Rules of Inference

1. **Modus Ponens:**

If $P \rightarrow Q$ and P is true, then Q must be true.

$$P \rightarrow Q, \quad P \quad \therefore Q$$

2. **Modus Tollens:**

If $P \rightarrow Q$ and Q is false, then P must be false.

$$P \rightarrow Q, \quad \neg Q \quad \therefore \neg P$$

3. **Disjunctive Syllogism:**

If $P \vee Q$ is true and $\neg P$ is true, then Q must be true.

$$P \vee Q, \quad \neg P \quad \therefore Q$$

Formal Proofs

Proofs in logic use a step-by-step application of inference rules to derive conclusions from premises. **Truth tables**, **natural deduction**, and **proof by contradiction** are common proof methods.

4. Applications in Mathematics and Computation

Mathematical Applications

- **Set theory:** Using logic to define and manipulate sets and their relationships.
- **Proof verification:** Ensuring mathematical theorems are logically sound.

- **Boolean algebra:** The foundation of logic gates in digital circuits.

Computational Applications

- **Artificial Intelligence:** Logical reasoning underpins knowledge representation and automated decision-making.
- **Formal Verification:** Checking the correctness of software and hardware.
- **Programming Languages:** Logic programming (e.g., Prolog) is based on declarative reasoning.

This foundational understanding of logic is essential for advanced topics in mathematical logic, theoretical computer science, and automated reasoning systems.